

**Homework Set 6**

(Due October 14, 2022 @ 12:00 pm in Gradescope and WileyPlus)

This assignment is to be individually. The number in parentheses next to each problem shows how many points the problem is worth in the overall assignment. Please show all work and use correct significant figures to receive full credit. Be sure to follow the problem formatting instructions to avoid unnecessary deductions. If you use an equation-solving tool (calculator, APEX, Solver, MATLAB), write out the equation(s) and note the tool that you used; otherwise, show all hand calculations used to solve any equations.

1. (25 pts) The goal of this problem is to use both Excel's Solver and APEX to solve the SRK equation of state to determine  $\hat{V}$  (L/mol) for CO<sub>2</sub> at (i) 200 K and 6.8 atm; (ii) 250 K and 12.3 atm; (iii) 300 K and 6.8 atm; (iv) 300 K and 21.5 atm; and (v) 300 K and 50.0 atm.

- (a) Starting with Equation 5.3-8, derive the following equivalent expression for the SRK equation of state:

$$f(\hat{V}) = P\hat{V}^3 - RT\hat{V}^2 + (a\alpha - b^2P - bRT)\hat{V} - aab = 0.$$

- (b) Prepare an Excel spreadsheet to take as inputs a species identifier (CO<sub>2</sub>), the critical temperature, critical pressure, Pitzer acentric factor, and the temperatures and pressures for which  $\hat{V}$  is calculated, and to calculate  $\hat{V}$  using Equations 5.3-8 to 5.3-13 for each of the specified conditions. The spreadsheet should have the following structure:

| SRK EQUATION SPREADSHEET |         |         |          |         |           |
|--------------------------|---------|---------|----------|---------|-----------|
| Species                  | CO2     |         |          |         |           |
| Tc (K)                   | *.***** |         |          |         |           |
| Pc (atm)                 | *.***** |         |          |         |           |
| omega                    | *.***** |         |          |         |           |
| a                        | *.***** |         |          |         |           |
| b                        | *.***** |         |          |         |           |
| m                        | *.***** |         |          |         |           |
|                          |         |         |          |         |           |
| T (K)                    | P (atm) | alpha   | V(ideal) | V (SRK) | f (V)     |
| 200                      | 6.8     | 1.3370  | 2.4135   | 2.1125  | *.***E-** |
| 250                      | 12.3    | *.***** | *.*****  | *.***** | *.***E-** |
| 300                      | 6.8     | *.***** | *.*****  | *.***** | *.***E-** |
| 300                      | 21.5    | *.***** | *.*****  | *.***** | *.***E-** |
| 300                      | 50.0    | *.***** | *.*****  | *.***** | *.***E-** |

**Include a screenshot of the completed Excel table when you submit part (b) in Gradescope and also submit your Excel spreadsheet in Moodle via the submission link. Name the Excel file "LastName\_FirstName\_HW6.xlsx".**

***Please note that any computer work submitted must be generated on your own personal computer or from your own NCSU Unity Account to avoid confusion about the origin of the file, and no sharing of files in any way is allowed.***

- (c) Use a root-finding procedure in a mathematical software package (e.g. APEX) to determine  $\hat{V}$  for each of the five conditions. Include a screenshot of the software package (e.g., APEX) output in your Gradescope submission for part (c).

2. **(20 pts)** An oxygen tank with a volume of  $2.5 \text{ ft}^3$  is kept in a room at  $50^\circ\text{F}$ . An engineer has used the ideal gas equation of state to determine that if the tank is first evacuated and then charged with  $35.3 \text{ lb}_m$  of pure oxygen, its rated *maximum allowable working pressure* (MAWP) will be attained. Operation at pressures above this value is considered unsafe.
- (a) What is the maximum allowable working pressure (psig) of the tank?
- (b) You suspect that at the conditions of the fully charged tank, ideal-gas behavior may not be a good assumption. Use the SRK equation of state to obtain a better estimate of the maximum mass of oxygen that may be charged into the tank. Did the ideal-gas assumption lead to a conservative estimate (on the safe side) or a non-conservative estimate of the amount of oxygen that could be charged?
- (c) Suppose the tank is charged and ruptures before the amount of oxygen calculated in Part (b) enters it. (It should have been able to withstand pressures up to four times the MAWP.) Think of at least five possible explanations for the failure of the tank below its rated pressure limit.
3. **(15 pts)** The product gas from a coal gasification plant consists of 60.0 mole% CO and the balance  $\text{H}_2$ ; it leaves the plant at  $150^\circ\text{C}$  and 135 bar absolute. The gas expands through a turbine, and the outlet gas from the turbine is fed to a boiler furnace at  $100^\circ\text{C}$  and 1 atm at a rate of  $425 \text{ m}^3/\text{min}$ . Estimate the inlet flow rate to the turbine in  $\text{ft}^3/\text{min}$ , using Kay's rule. What percentage error would result from the use of the ideal-gas equation of state at the turbine inlet?

**(40 pts)** The remainder of the assignment (4 problems) will be completed individually online using WileyPLUS. *You do not have to turn in any paperwork with this portion of the assignment.* Use the link for your class on the Moodle site, and then you can access the Assignment within WileyPLUS. The due date for the WileyPLUS completion is the same as for the homework assignment – the beginning of your class period. Note that the WileyPLUS assignment cannot be submitted late.

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### Honors Contract Problem

Determining the value of newly located natural gas sites involves estimating the gas composition, quantity, and ease of access. For example, one report described a find of 2 trillion cubic feet of natural gas that is significantly offshore, in 20 feet of water, and at a drilled depth of 25,000 ft. (In North America and the OPEC countries, reported volumes are determined at 14.73 psia and  $60^\circ\text{F}$ .) The pressure in this find is estimated to be 750 atm, and the gas is 94 mole% methane, 3.5% ethane, and the balance  $\text{CO}_2$ .

- (a) Estimate the total lb-moles of gas in the find.
- (b) Use the compressibility-factor equation of state to estimate the specific volume ( $\text{ft}^3/\text{lb-mole}$ ) in the well. The temperature of such wells can vary depending upon a number of factors; for the purposes of this problem, assume that it is  $200^\circ\text{C}$ . State any additional assumption that you need to make for the mixture.